Lecture 5: Practice Problems

Math for Machine Learning

October 24, 2026

1 Ordinary Differential Equations

Problem 1.1 (**Mandatory**)**.** (a) Solve $x' = -2x$ with $x(0) = 3$. Compute $x(1)$ and the time after which x has halved.

(b) Show that $x(t) = \frac{1}{1+4e^{-t}}$ solves the logistic equation $x' = x(1-x)$ with $x(0) = 0.2$, and write it with the sigmoid function. Compute $x(2)$ and $\lim_{t\to\infty} x(t)$.

Problem 1.2. (a) Show that $x(t) = \frac{1}{1-t}$ solves $x' = x^2$ with $x(0) = 1$. What happens as $t \to 1$? Why does this not contradict the existence and uniqueness theorem?

(b) Show that $x(t) = 0$ and $x(t) = t^2$ both solve $x' = 2\sqrt{|x|}$ for $t \geq 0$, with $x(0) = 0$. Which assumption of the theorem fails?

2 Linear Systems

Problem 2.1 (**Mandatory**)**.** Let $A = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$.

(a) Find the eigenvalues and eigenvectors of A, and solve $\mathbf{x}' = A\mathbf{x}$ with $\mathbf{x}(0) = (1, 0)^\top$. Compute $\mathbf{x}(1)$.

(b) For which initial values does the solution tend to $\mathbf{0}$? Sketch the solutions. (The origin is a saddle point of this ODE.)

(c) Compute e^{At}.

Problem 2.2 ($\star$)**.** Let $A = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$ with $\omega > 0$.

(a) Show that the solutions of $\mathbf{x}' = A\mathbf{x}$ move on circles with angular speed ω. What is the period?

(b) Show that Euler's method satisfies $\|\mathbf{x}_{k+1}\|^2 = (1 + h^2\omega^2)\|\mathbf{x}_k\|^2$, so the numerical solution spirals outwards. For $\omega = 1$ and $h = 0.1$, by which factor does $\|\mathbf{x}_k\|$ grow during one period?

3 Gradient Flow and Euler's Method

Problem 3.1 (Mandatory). Let $f(x_1, x_2) = x_1^2 + 4x_2^2$.

(a) Solve the gradient flow $\mathbf{x}' = -\nabla f(\mathbf{x})$ with $\mathbf{x}(0) = (2, 1)^\top$, and check that $f(\mathbf{x}(t))$ decreases.

(b) Write down Euler's method for this flow with step h, and compare it with gradient descent. For which h do the iterates converge to $\mathbf{0}$? What happens for $h = 0.2$ and $h = 0.3$?

Problem 3.2. Apply Euler's method to $x' = -x$, $x(0) = 1$ on the interval $[0, 1]$, with $h = 0.25$ and $h = 0.125$. Compare with $x(1) = e^{-1}$. By which factor does the error shrink when h is halved?

4 Flows, Volumes and Densities

Problem 4.1 (Mandatory). (a) For the ODE $\mathbf{x}' = A\mathbf{x}$ with $A = \begin{pmatrix} 1 & 2 \\ 0 & -3 \end{pmatrix}$, compute the divergence of the vector field and the factor by which the flow changes areas in the time t. Check the result with the eigenvalues of A.

(b) Let $F(x, y) = (x - x^3, -y)^\top$. Compute $\operatorname{div} F$. Find the points where $F = \mathbf{0}$, and the divergence there. Where does the flow preserve areas to first order, and where does it compress them?

Problem 4.2 ($\star$). Let $x' = -x$, and let $x(0)$ be random with the standard normal density φ.

(a) Show that $x(t) = e^{-t}x(0)$ has the density $p_t(x) = e^t\varphi(e^t x)$.

(b) Verify the continuity equation $\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x}(p_t F)$ for $F(x) = -x$.

(c) Show that $\log p_t(x(t))$ increases at the rate $1 = -\operatorname{div} F$ along each solution, and explain this with the volume interpretation of the divergence.

5 Stochastic Differential Equations

Problem 5.1 (Mandatory). Let $B(t)$ be a Brownian motion.

(a) Find the distribution of $B(4) - B(1)$, and $P(B(4) > 2)$.

(b) Compute $\operatorname{Corr}(B(1), B(2))$ and the distribution of $B(1) + B(2)$. (Hint: write $B(2) = B(1) + (B(2) - B(1))$.)

(c) Given $B(1) = 1$, what is the distribution of $B(2)$? Given $B(2) = 2$, what is the distribution of $B(1)$? (Use the conditioning theorem for Gaussians.)

Problem 5.2 (Mandatory). Consider the SDE $dX = -2X\,dt + 0.5\,dB$ with $X(0) = 1$.

(a) Perform two Euler–Maruyama steps with $h = 0.1$, using the standard normal samples $Z_0 = 0.5$ and $Z_1 = -1$.

(b) Find $\mathbb{E}[X_k]$ for the Euler–Maruyama chain, and compare $\mathbb{E}[X_{10}]$ with the mean e^{-2t} of the SDE at $t = 1$.

(c) Find the stationary variance of the SDE, and of the chain with $h = 0.1$. For which step sizes does the chain have a stationary distribution?

Problem 5.3. (Why $\sqrt{h}$?) Divide $[0, t]$ into $n = t/h$ steps, and consider a scheme that adds the noise $\sigma h^\gamma Z_k$ in step k, with i.i.d. $Z_k \sim \mathcal{N}(0, 1)$. Compute the variance of the total noise $\sigma h^\gamma (Z_0 + \cdots + Z_{n-1})$, and show that as $h \to 0$, it tends to 0 for $\gamma > \frac{1}{2}$ and to ∞ for $\gamma < \frac{1}{2}$, while it equals $\sigma^2 t$ for $\gamma = \frac{1}{2}$.

Problem 5.4 ($\star$)**.** (Quadratic variation.) Divide $[0, T]$ into n intervals of the length $h = T/n$, and let

$$Q_n = \sum_{k=0}^{n-1} \big(B((k+1)h) - B(kh)\big)^2.$$

(a) Using $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathcal{N}(0, 1)$, show that $\mathbb{E}[Q_n] = T$ and $\mathrm{Var}(Q_n) = 2Th$, so that $Q_n \to T$ as $n \to \infty$.

(b) Show that for a continuously differentiable function f on $[0, T]$, the sum $\sum_k \big(f((k+1)h) - f(kh)\big)^2$ tends to 0. (Hint: mean value theorem.) Conclude that Brownian paths are not continuously differentiable.

(This is the rule "$(dB)^2 = dt$" of stochastic calculus.)

6 Langevin Dynamics and Diffusion Models

Problem 6.1 (**Mandatory**)**.** (Langevin sampling of a Gaussian.) Let $U(x) = \frac{(x-\mu)^2}{2s^2}$, so that e^{-U} is proportional to the density of $\mathcal{N}(\mu, s^2)$.

(a) Write down the unadjusted Langevin algorithm for U with the step h, and show that it has the form $x_{k+1} - \mu = a(x_k - \mu) + \sqrt{2h}\, z_k$.

(b) Show that for $0 < h < 2s^2$, the chain has the stationary distribution $\mathcal{N}\big(\mu, \frac{2s^2}{2 - h/s^2}\big)$. Evaluate the variance for $\mu = 2$, $s^2 = 0.25$ and $h = 0.01$.

(c) What does the algorithm do without the noise term? How is this related to MAP estimation?

Problem 6.2 ($\star$)**.** (The forward process of a diffusion model.) A data point $x_0 = 3$ is gradually noised by $x_t = \sqrt{1-\beta}\, x_{t-1} + \sqrt{\beta}\, z_t$ for $t = 1, 2, \ldots$, with i.i.d. $z_t \sim \mathcal{N}(0, 1)$ and $\beta = 0.02$.

(a) Show by induction that $x_t \sim \mathcal{N}\big((1-\beta)^{t/2} x_0,\ 1 - (1-\beta)^t\big)$. Why is this useful for training, where x_t is needed for random times t?

(b) Compute the mean and the variance for $t = 10$, 100 and 500. What is the distribution of x_t for large t, whatever the data point?

(c) Show that the process is approximately the Euler–Maruyama scheme with $h = \beta$ for the Ornstein–Uhlenbeck SDE $dX = -\frac{1}{2} X\, dt + dB$. (Hint: $\sqrt{1-\beta} \approx 1 - \frac{\beta}{2}$.)